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***B.Tech. Degree V Semester Examination November 2014*****SE 1505 MANUFACTURING PROCESS**  
(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

**PART A**  
(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) How are engineering materials classified?  
(b) Draw iron-carbon diagram and specify its regions.  
(c) What are the common weld defects?  
(d) What are the methods adopted for the inspection of welded joints?  
(e) Write desirable properties of moulding sand.  
(f) What is die casting?  
(g) Explain abrasive jet machining.  
(h) Write the different kind of operations that can be performed on a lathe.

**PART B**

(4 × 15 = 60)

- II. Explain some of the important non-ferrous metals, its properties and uses.  
**OR**  
III. Describe the purpose and process of (i) annealing and (ii) hardening  
IV. Explain arc welding in detail.  
**OR**  
V. Write notes on:  
(i) electron beam welding  
(ii) laser beam welding  
VI. Describe different types of allowances provided on pattern.  
**OR**  
VII. Write notes on:  
(i) centrifugal casting  
(ii) investment casting  
VIII. Differentiate between hot working and cold working of metals.  
**OR**  
IX. Explain orthogonal cutting and oblique cutting, with neat sketch. What are the common cutting tool materials?

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